

Two-Step Equations

AQA · KS3 Year 8

Name: _____ Class: _____ Date: _____

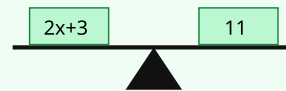
Solve linear equations of the form $ax + b = c$ that need two steps. Undo the addition or subtraction first, then undo the multiplication or division, keeping the equation balanced.

VISUAL EXAMPLE

$$2x + 3 = 11$$

$$-3 \text{ both sides} \quad 2x = 8$$

$$\div 2 \text{ both sides} \quad x = 4$$



An equation balances: do the same to both sides to keep it level.

COMMON MISTAKES

Dividing before undoing the + or - — undo the addition/subtraction first.

Forgetting to do each step to both sides.

Dividing only the number, not the whole side: $2x = 8$ gives $x = 4$, not $2x = 4$.

Losing a minus sign when the constant is negative.

METHOD

- 1 Identify the term with the letter and the number added or subtracted.
- 2 Undo the + or - by doing the inverse to both sides.
- 3 You now have a one-step equation like $ax = c$.
- 4 Undo the $\times$ or $\div$ by doing the inverse to both sides.
- 5 Check by substituting your answer into the original equation.

WORKED EXAMPLE

Solve: $2x + 3 = 11$

Step 1 — The letter term is $2x$; 3 has been added.

Step 2 — Subtract 3 from both sides: $2x = 8$.

Step 3 — One-step equation: $2x = 8$.

Step 4 — Divide both sides by 2: $x = 4$.

Step 5 — Check: $2 \times 4 + 3 = 11$

1 Solve.

(2 marks)

a) $2x + 1 = 7$

b) $3x + 4 = 19$

HOW TO START

Undo the number that is added or subtracted FIRST.

For $2x + 1 = 7$, subtract 1: $2x = 6$. Then divide by 2: $x = 3$.

ANSWERS

(a) = _____

(b) = _____

METHOD 1 Spot the letter term and the + / - number | 2 Undo the + or - first (both sides) | 3 Now it is a one-step equation | 4 Undo the $\times$ or $\div$ (both sides) | 5 Check by substituting

2 Solve.

(2 marks)

a) $5x - 2 = 13$

b) $4x - 7 = 9$

HOW TO START

Same order: undo the - first, then the $\times$.

For $5x - 2 = 13$, add 2: $5x = 15$. Then divide by 5: $x = 3$.

ANSWERS

(a) = _____

(b) = _____

METHOD 1 Spot the letter term and the + / - number | 2 Undo the + or - first (both sides) | 3 Now it is a one-step equation | 4 Undo the $\times$ or $\div$ (both sides) | 5 Check by substituting

3 Solve.

(2 marks)

a) $x \div 2 + 3 = 8$

b) $x \div 3 - 1 = 4$

ANSWERS

(a) = _____

(b) = _____

METHOD 1 Spot the letter term and the + / - number | 2 Undo the + or - first (both sides) | 3 Now it is a one-step equation | 4 Undo the $\times$ or $\div$ (both sides) | 5 Check by substituting

4 Solve. Take care with the negatives.

(3 marks)

a) $2x + 9 = 3$

b) $5x + 14 = 4$

c) $3x - 5 = -11$

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Spot the letter term and the + / - number | 2 Undo the + or - first (both sides) | 3 Now it is a one-step equation | 4 Undo the $\times$ or $\div$ (both sides) | 5 Check by substituting

5 Solve.

(3 marks)

a) $7 = 2x + 1$

b) $10 = 4x - 6$

c) $(x + 5) \div 2 = 6$

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Spot the letter term and the + / - number | 2 Undo the + or - first (both sides) | 3 Now it is a one-step equation | 4 Undo the $\times$ or $\div$ (both sides) | 5 Check by substituting

(3 marks)

b) I think of a number, multiply it by 3, then subtract 5. The result is 16. Find the number.

(b) =

METHOD 1 Spot the letter term and the $+$ / $-$ number | 2 Undo the $+$ or $-$ first (both sides) | 3 Now it is a one-step equation | 4 Undo the $\times$ or $\div$ (both sides) | 5 Check by substituting

7 A rectangle has length $(2x + 1)$ cm and width 4 cm.
The perimeter is 30 cm.

(3 marks)

- a) Form an equation in x . (1 mark)
- b) Solve it to find x . (1 mark)
- c) Work out the length of the rectangle. (1 mark)

ANSWERS

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Spot the letter term and the $+$ / $-$ number | 2 Undo the $+$ or $-$ first (both sides) | 3 Now it is a one-step equation | 4 Undo the $\times$ or $\div$ (both sides) | 5 Check by substituting

(5 marks)

- Form an equation and find the hourly charge c . (2 marks)
- Work out the cost of a 5-hour job. (1 mark)
- Another job cost £75. How many hours did it take? (2 marks)

(a) = _____

(b) = _____

(c) = _____

METHOD 1 Spot the letter term and the $+$ / $-$ number | 2 Undo the $+$ or $-$ first (both sides) | 3 Now it is a one-step equation | 4 Undo the $\times$ or $\div$ (both sides) | 5 Check by substituting

Self-reflection

How confident do you feel with this topic now? (tick one)

Not yet

Getting there

Confident

Really confident

Which question did you find the hardest, and why?

Why do we undo the + or - before the $\times$ or $\div$?

One thing I will practise before the next lesson:

Teacher key

Q1: a) $2x = 6$ $x = 3$ b) $3x = 15$ $x = 5$

Q2: a) $5x = 15$ $x = 3$ b) $4x = 16$ $x = 4$

Q3: a) $x \div 2 = 5$ $x = 10$ b) $x \div 3 = 5$ $x = 15$

Q4: a) $2x = -6$ $x = -3$ b) $5x = -10$ $x = -2$ c) $3x = -6$ $x = -2$

Q5: a) $2x = 6$ $x = 3$ b) $4x = 16$ $x = 4$ c) $x + 5 = 12$ $x = 7$

Q6: a) $2m + 3 = 17$ $2m = 14$ $m = 7$ miles b) $3n - 5 = 16$ $3n = 21$ $n = 7$

Q7: a) $2(2x + 1 + 4) = 30$, i.e. $4x + 10 = 30$ b) $4x = 20$ $x = 5$ c) length = $2(5) + 1 = 11$ cm

Q8: a) $15 + 3c = 60$ $3c = 45$ $c = \text{£}15/\text{hour}$ b) $15 + 5 \times 15 = \text{£}90$ c) $15 + 15h = 75$ $15h = 60$ $h = 4$ hours
